Vijay Raj Venugopal

Mechanical Engineer

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PROFESSIONAL SUMMARY

Mechanical Engineering graduate with hands-on experience in field engineering, infrastructure projects, facilities operations, and system integration. Proven ability to handle fast-paced, high-pressure environments while maintaining safety, quality, and operational efficiency. Experienced in supervising on-site teams, improving process reliability, resolving technical issues, and supporting cross-functional engineering efforts. Skilled in simulation, system testing, and data analysis, with practical exposure to robotics and software integration. Demonstrated success in optimizing workflows and enhancing project outcomes. Seeking an entry-level mechanical or manufacturing engineering role to apply technical expertise, problem-solving ability, and a results-driven mindset in a dynamic engineering environment.

EDUCATION

Master Of Engineering in Mechanical Engineering GPA: 3.54
Old Dominion University, Norfolk, VA, US December 2025

Bachelor Of Engineering in Mechanical Engineering GPA: 3.33
Bir Tikendrajit University, India December 2023

Java Certification Course
Quest Innovative Solutions Pvt. Ltd. April 2023

WORK EXPERIENCE

Old Dominion University November 2024 – December 2025
Facilities Management – Part Time

- Managed and closed a high volume of facility work orders in a fast-paced university setting, ensuring timely completion and minimal disruption.
- Coordinated with cross-functional teams to maintain safety standards, improve task efficiency, and support smooth daily operations.
- Strengthened organizational and time-management skills by prioritizing multiple requests and deadlines across campus facilities.
- Delivered professional, courteous service when addressing facility needs for faculty, staff, and students, reinforcing a positive campus experience.

STRIA Constructions October 2021 – November 2022
Engineer Trainee

- Supported the Quality Control department by preparing routine progress reports and final handover documentation for clients, ensuring accuracy and completeness.
- Organized and maintained project documentation and administrative records to keep quality and construction files up to date.

- Monitored gas pipeline construction activities, including trenching, field joint coating, crossings, lowering, and backfilling, to track progress and identify issues early.
- Verified compliance with approved procedures, applicable codes, and construction drawings during site activities.
- Built strong proficiency in quality control practices, documentation management, and adherence to industry standards through hands-on field and office experience.

SAS Electricals June 2020 – October 2021
September 2023 – December 2023

Site Engineer

- Supervised the installation and alignment of water pipelines for the Kerala Water Authority, ensuring work followed project plans and specifications.
- Measured trench depth and monitored excavation activities to maintain compliance with safety regulations and minimize site risks.
- Verified pipe batch numbers, inspected materials, and maintained quality control throughout pipeline installation.
- Coordinated site crews, assigned daily tasks, and managed work schedules to keep construction activities on track.
- Ensured the availability and proper allocation of tools, machinery, and materials to support uninterrupted workflow and timely project progress.
- Demonstrated effective site management, strong safety compliance, and collaborative teamwork across multiple infrastructure projects.

Quest Innovative Solutions Pvt Ltd January 2023 – October 2023
Trainee Software Engineer - Internship

- Completed an intensive Java Full Stack Development training program (January 9–April 24, 2023), building a strong foundation in core Java and web technologies.
- Continued as a Trainee Software Engineer from April 25–October 25, 2023, applying training knowledge to real project work.
- Gained hands-on experience in full-stack development, including front-end interfaces and back-end services using Java-based technologies.
- Contributed to software development, debugging, and code optimization to improve application performance and reliability.
- Collaborated with team members to design, develop, and implement software solutions that met client and project requirements.
- Developed solid proficiency in Java programming, database management, and full-stack application deployment through practical, project-driven work.

SKILLS AND LANGUAGES

Technical Skills

- **Simulation & CAD:** AutoCAD, Gazebo, RViz2
- **Programming & Software:** Python, Java, MATLAB, Linux (Ubuntu)
- **Robotics & Autonomy:** ROS2, SLAM (Cartographer), Navigation2, Localization, Mapping, Path Planning (A*, Dijkstra), Sensor Fusion

- **Data Analysis & Documentation:** Microsoft Word, Excel, PowerPoint
- **Sensors:** LiDAR, Camera, Ultrasonic Sensors

Soft Skills

- Problem solving
- Team collaboration
- Adaptability
- Time management
- Conflict resolution
- Stakeholder and customer focus
- Communication (verbal and written)
- Leadership and mentoring
- Planning and prioritization
- Attention to detail
- Initiative and ownership
- Critical thinking and decision-making

Languages

- English – Full professional proficiency
- Malayalam – Native / Bilingual proficiency
- Tamil – Conversational
- Hindi – Basic proficiency

PROJECT EXPERIENCE

Autonomous Indoor Robot System Integration Using ROS2, SLAM, and Multi-Sensor Perception Master's Project | 2025

- Designed and integrated an autonomous indoor mobile robot using ROS2 on Raspberry Pi–based platforms for real-time navigation and control.
- Implemented SLAM (Cartographer) for real-time mapping and localization in complex indoor environments.
- Integrated LiDAR, camera, and ultrasonic sensors to enable multi-sensor perception and reliable obstacle detection.
- Configured and deployed the Navigation2 (Nav2) stack to achieve autonomous waypoint navigation and path execution.
- Tuned costmaps and planners (A* / Dijkstra) to ensure safe navigation in constrained indoor spaces.
- Validated navigation performance across multiple test environments, achieving repeatable localization accuracy.
- Built and verified TF trees and sensor frames to maintain consistent coordinate transforms and system stability.
- Performed iterative system testing, diagnosing hardware–software integration issues and improving navigation reliability.
- Analyzed experimental data and documented results using Excel and technical reports to support design decisions.
- Tested and verified system performance in both real-world indoor trials and Gazebo simulations.
- Documented system architecture, experiments, and key performance metrics for the final Master's report and presentation.

Study on solar combined with A/C assisted water heat pump Undergraduate Project | 2019

- Investigated ways to improve energy efficiency and reduce environmental impact by recovering waste heat from split air conditioners.

- Designed a heat recovery system to capture condenser heat and repurpose it for domestic water heating.
- Integrated a solar collector to boost heat absorption and maximize overall energy utilization.
- Developed a techno-economically feasible concept aimed at lowering energy costs and minimizing resource misuse.
- Conducted an in-depth analysis of heat pump performance, waste heat recovery potential, and sustainable energy solutions.

LEADERSHIP

Rise Up Forum July 2019 – December 2023

- Volunteered in multiple community relief initiatives, providing humanitarian assistance and disaster relief to affected populations.
- Supported on-ground coordination, distribution of essential supplies, and outreach efforts as part of a dedicated volunteer team.

Y's Men March 2016-March 2022

- Active member of a community service club focused on local outreach and social impact initiatives.
- Served as President of the Y's Youth Club for 2 years, leading projects and events, after 6 years of continuous membership and involvement.

MRA

- Volunteered in a range of charitable and relief initiatives, supporting communities in need through organized service activities.